

Deep Learning Provides a Full Image of Brain Function and Adaptability from the Surface

Ali Alattar, Karam Khateeb, Yazdan Shahmorad

Abstract

Modern neuroscience seeks to link large-scale brain dynamics with the activity of individual neurons, but noninvasive recordings lack cellular resolution. Broad-field techniques like EEG and ECoG capture global patterns of neural activity yet cannot pinpoint which neurons fire, obscuring fine circuit details. This proposal addresses that critical gap by leveraging deep learning to infer single-neuron spike information from surface electrophysiology. We will develop a novel data-driven framework that deconvolves aggregate surface signals to recover spike-level events, enabling high-resolution neural readouts without invasive electrodes. Using this methodology, we will monitor how targeted interventions, such as neurofeedback training or stimulation, induce plastic changes in specific neural circuits over time. Our approach is innovative: by combining noninvasive recordings with advanced AI-based spike sorting, it bridges the scale divide between macroscopic brain rhythms and microscopic neuronal activity. If successful, this approach will provide a powerful new window into human brain function and adaptability, with far-reaching impact on basic neuroscience, neuroengineering, and clinical neurotechnology.

Introduction

Understanding brain function and plasticity requires recording neural activity across large networks over long timescales. Multichannel electrophysiology has become essential for decoding neural connectivity and behavior¹. Noninvasive or surface-level methods such as electroencephalography (EEG) and electrocorticography (ECoG) can simultaneously capture broad patterns of cortical activity (local field potentials, LFPs) across multiple areas, enabling human brain mapping without penetrating electrodes^{1–3}. However, surface electrodes are relatively distant from neurons and have large contact areas, which blurs spatial detail^{1,2}. Fig. 1 illustrates this limitation: intracortical microelectrodes record individual neuron spikes at high temporal resolution, whereas each surface ECoG electrode samples the summed signals of millions of neurons. Consequently, surface recordings provide excellent coverage but poor single-neuron resolution^{1,3}. Recent work with high-density ECoG grids has shown that some high-frequency multiunit activity can be detected on the cortical surface³, yet these signals still represent the combined firing of many neurons rather than isolated spikes. Surface LFPs mainly reflect the synchronized synaptic currents from local populations (on the order of a few hundred microns around each electrode)¹, while action potentials from individual neurons are high-frequency transients observable only with electrodes very close to those cells^{1,4}. This disparity poses a fundamental challenge: large-scale surface recordings cannot directly reveal which neurons are firing, making it difficult to resolve fine-grained neural communication or track plastic changes in specific circuits.

Traditional electrophysiology approaches tackle this problem by using high-density intracortical arrays and spike-sorting algorithms to separate each neuron's spikes from raw extracellular signals^{4–6}. Indeed, the latest silicon probes can record from hundreds of neurons simultaneously, providing unprecedented data for network analysis⁴. However, extracting single-neuron spike trains from such data is nontrivial: each electrode typically picks up spikes from multiple nearby cells plus noise^{4,5}. Spike sorting remains an active area of research^{5,6}, and even state-of-the-art methods face limitations in accuracy, scalability, and reliability^{5,6}. Overlapping spikes (when multiple neurons fire at once) and low signal-to-noise ratios often cause sorting errors⁵. Moreover, conventional spike-sorting pipelines rely on manual feature selection, such as waveform amplitude or principal components, that may not capture subtle differences between spike waveforms^{5,6}. Principal component analysis (PCA) has long been used to reduce spike waveform dimensionality for clustering^{5,6}, forming the backbone of many sorting tools^{4–6}. Yet standard techniques optimized for clean microelectrode recordings struggle on noisier, lower-amplitude surface signals^{1–4}. In practice, surface EEG/ECoG recordings are usually analyzed only at the population level (oscillatory rhythms, spectral power) rather than for spike content, because distinguishing single spikes in such data is extremely difficult with classical methods^{4–6}.

Despite these challenges, there are intriguing hints that surface-level signals carry latent information about individual neuron firing. For instance, high-frequency “gamma” LFP activity correlates strongly with local spiking rates⁷. In intracortical recordings, LFP power in the ~60–200 Hz band (overlapping the spike frequency range) tracks the firing of nearby neurons⁷. Similarly, human ECoG studies have shown that broadband high-frequency signals reflect the summed firing of local neuronal populations⁸. Recent studies even demonstrate that spikes can be algorithmically inferred from LFP signals using machine learning^{9–13}. For example, Ahmadi et al. reconstructed aggregate spiking activity from low-frequency intracortical LFPs, confirming that these field potentials contain considerable information about underlying spikes⁹. Likewise, Hong et al. applied deep learning to recover high-frequency spike waveforms from intentionally down-sampled recordings¹⁰. These advances suggest it may be possible to deconvolve or disentangle single-neuron events from large-scale surface signals, bridging the gap between noninvasive recordings and cellular-level data^{9–13}. However, no existing framework fully resolves individual spikes from truly surface-recorded signals, like ECoG or scalp EEG, in a robust and generalizable manner.

Here, we propose to develop a novel methodology that leverages deep learning to extract individual neuronal spike information from surface-level brain recordings and to use this capability to reveal how targeted interventions can shape brain network function and adaptability over time. Fig. 1 outlines the concept: by enhancing the temporal and spatial resolution of surface EEG/ECoG signals, this approach will uncover fine-grained neural interactions normally hidden in the summed activity^{1, 2, 8–13}. Our approach will combine data-driven feature learning with neural signal processing to isolate spike-like patterns from noninvasive recordings. Using this tool, we will observe how intentional perturbations, such as neurofeedback training or noninvasive brain stimulation, induce changes in specific neural circuits. This will provide new insights into neuroplasticity that were previously accessible only with invasive methods^{14, 19}. Unlike prior work that either relies on penetrating electrodes or treats surface signals as purely macroscopic measures, our project will extract single-neuron information from surface recordings for the first time and link system-level interventions to microscopic neural adaptations^{1, 9, 13, 19}. By uniting deep learning-based spike inference with longitudinal brain modulation experiments, we will create a powerful new framework to study brain function and plasticity noninvasively^{10, 13, 19, 23}.

Specific Aim 1: Isolate spike-level patterns from surface recordings via PCA decomposition.

We will explore whether principal component analysis of multi-channel surface ECoG/EEG signals can separate meaningful components corresponding to individual neuronal spiking activity^{5, 6}. In this aim, neural recordings will be filtered to emphasize high-frequency content (e.g. 300–500 Hz) that could contain faint spike signatures^{1, 7}. We will then apply PCA across channels and time windows to identify basis components that capture recurring sharp-wave waveforms⁵. The rationale is that while single spikes are not directly visible in low-amplitude surface traces, they may manifest as subtle covariance patterns across electrodes. By projecting the data into PCA space, we hope to reveal weak spike-like signals distinguishable from background noise and slower oscillations^{6, 7}. Prior work has shown that PCA is effective in capturing spike waveform variance in invasive recordings^{5, 6}. Here we will test its capability on surface-level data, using ground-truth comparisons when possible (for example, simultaneous microelectrode recordings in an animal model or hybrid ECoG-depth electrodes in patients)^{3, 8}. We anticipate that a few principal components will resemble canonical spike shapes or templates, indicating the presence of unit-specific activity⁶. Success in Aim 1 will provide preliminary confirmation that surface signals contain latent single-spike information beyond what standard LFP analyses reveal^{1, 9}. It will also inform the feature space for subsequent aims. Fig. 2 will illustrate this approach, showing raw surface signals decomposed into principal components that highlight candidate spike events.

Specific Aim 2: Improve single-neuron spike resolution using threshold detection, clustering, and machine learning.

We will extract candidate spike events from the surface recordings and refine their separation into putative neuronal sources. First, we will perform threshold-based spike detection on the band-passed signals, a common step in conventional spike sorting^{4, 5}. Each threshold crossing will be treated as a “spike snippet” waveform. Because many such snippets will be mixtures of multiple neurons or noise, we will employ a divide-and-conquer clustering strategy to organize them¹¹. Specifically, we will group detected spikes initially by the electrode on which the largest amplitude occurs, similar to methods used for multi-electrode array sorting^{6, 11}. Within each initial group, we will further cluster the spike snippets based on waveform features (for example,

the principal-component scores from Aim 1) to segregate distinct candidate units^{5, 6}. We will then apply machine learning classifiers to these clustered spike waveforms to improve identification of unique neurons¹². For instance, we will train a convolutional neural network (CNN) or gradient-boosting classifier on the PCA-reduced spike snippets to automatically distinguish putative neuron clusters¹². Recent advances in spike sorting have shown that trained classifiers can outperform manual clustering, achieving high accuracy even when spikes from different neurons overlap in time^{12, 13}. By training on an initial batch of hand-labeled or simulated spikes with known ground truth, the classifier will learn to assign new spike events to the correct neuron category¹². We expect this approach to sharpen the sorting of surface-recorded spikes, yielding clearer single-neuron separation than Aim 1 alone. Fig. 2 will illustrate this processing pipeline.

Specific Aim 3: Develop a self-supervised deep learning model to learn spike features directly from surface data.

The goal of Aim 3 is to move beyond manual feature engineering (PCA, thresholding) and instead train a deep neural network to directly discover and disentangle spike signatures in surface signals. We will design a self-supervised learning framework in which the model learns from unlabeled recordings. One approach will use an autoencoder or contrastive learning scheme: the network's encoder takes raw multi-channel ECoG/EEG segments as input and is trained to produce a low-dimensional embedding that represents the presence of different spike types, without requiring explicit labels^{21, 22}. We will incorporate techniques recently introduced in the spike-sorting field, such as pseudo-labeling and iterative refinement^{13, 22}. For example, the model could first cluster the input signals in an unsupervised manner to generate provisional spike labels (using the Aim 2 output as an initialization). Those labels are then used to fine-tune the network in a supervised fashion, and the process repeats – similar to the approach of Brockhoff et al. in PseudoSorter¹³. The network architecture will likely involve temporal convolutional layers to capture spike waveforms and a latent embedding space that separates distinct neuron signatures^{21–23}. Fig. 3 will depict this model, including the encoder–decoder structure and the iterative training loop. We will validate the deep model's performance against the results of Aim 2 and any available ground-truth data^{9, 10, 13}. Finally, we will deploy it in a longitudinal experiment to track how intentional interventions influence neural activity. By applying our model to recordings before, during, and after an intervention (for example, a neurofeedback training regimen), we will examine changes in inferred spike patterns and connectivity using only surface EEG/ECoG data^{14, 19}. A successful Aim 3 demonstration will show that our deep learning pipeline can noninvasively monitor neuron-level plasticity – for example, detecting increases in the coordinated firing of a circuit after targeted training – which was not previously possible with surface recordings alone.

Impact

Our project will advance basic neuroscience by bridging macroscopic brain signals and microscopic neuronal activity, enabling noninvasive studies of fine-scale neural dynamics in humans and animal models over long timescales^{1, 4, 9}. By extracting single-neuron information from surface recordings, we aim to achieve with noninvasive EEG/ECoG some of the cellular-level insights previously limited to invasive arrays like Neuropixels^{15, 16}. This approach could also enhance brain–machine interfaces (BMIs) by improving decoding of individual neuron signals from surface data, leading to more precise and adaptive BMI algorithms^{12, 17, 24}. Clinically, inferring spike-level events from ECoG or EEG could boost performance in BCIs for patients and refine epilepsy mapping or neurostimulation interventions²⁴, while also serving as a sensitive biomarker for disease monitoring and rehabilitation tracking^{14, 17, 25}. By reducing reliance on invasive recordings, our methods will make neuroscience and neurotechnology safer, more accessible, and more cost-effective, aligning with broader goals of improving healthcare and human productivity through brain research and engineering^{17, 24, 25}.

Figures

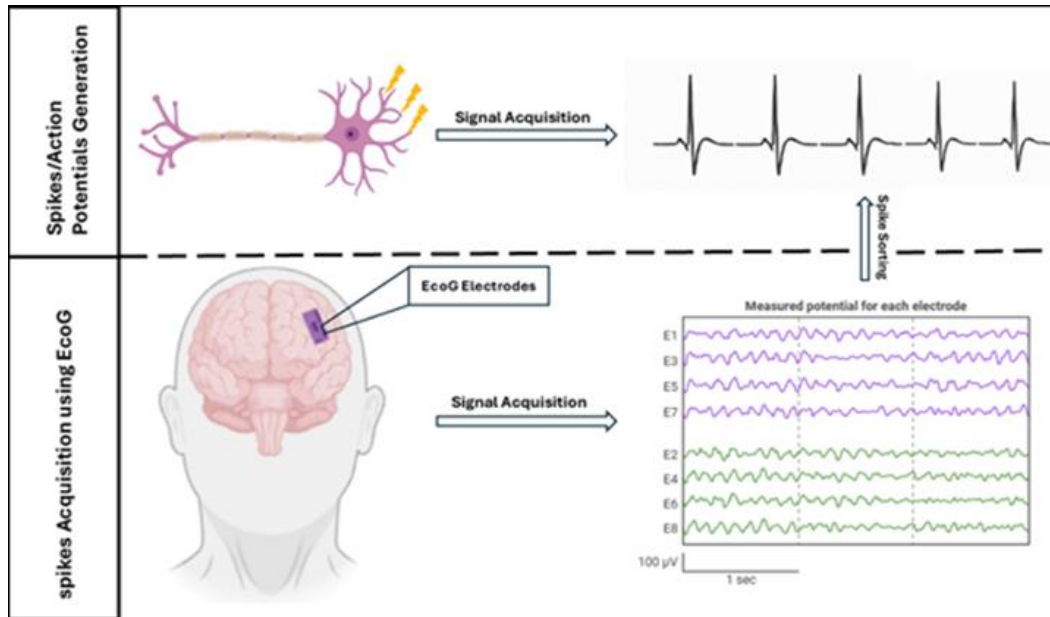


Figure 1. Direct Neuronal vs ECoG Spike Acquisition. (Top) Neuronal action potentials are recorded directly as spike trains via intracellular methods. (Bottom) ECoG electrodes on the cortical surface capture large-scale field potentials; spike sorting is applied to extract individual spikes. The figure highlights the differences in signal detail between intracellular and surface acquisition methods.

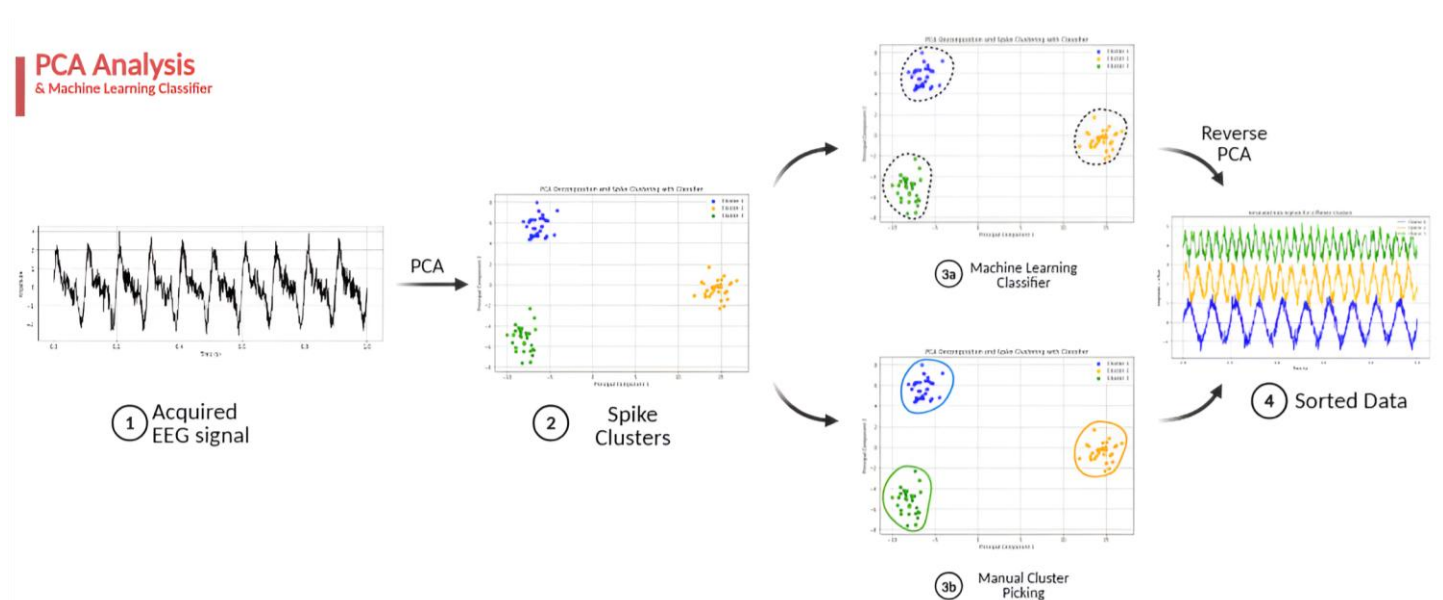


Figure 2. PCA decomposition and spike clustering pipeline. Raw ECoG/LFP signals are band-pass filtered and decomposed via PCA to extract principal components that resemble neuronal spikes. Detected spike events (threshold crossings) are grouped by their channel of origin and then clustered based on waveform features. A machine learning classifier is finally applied to these clusters to refine single-neuron spike resolution. (This figure will schematically visualize the processes of Aim 1 and Aim 2.)

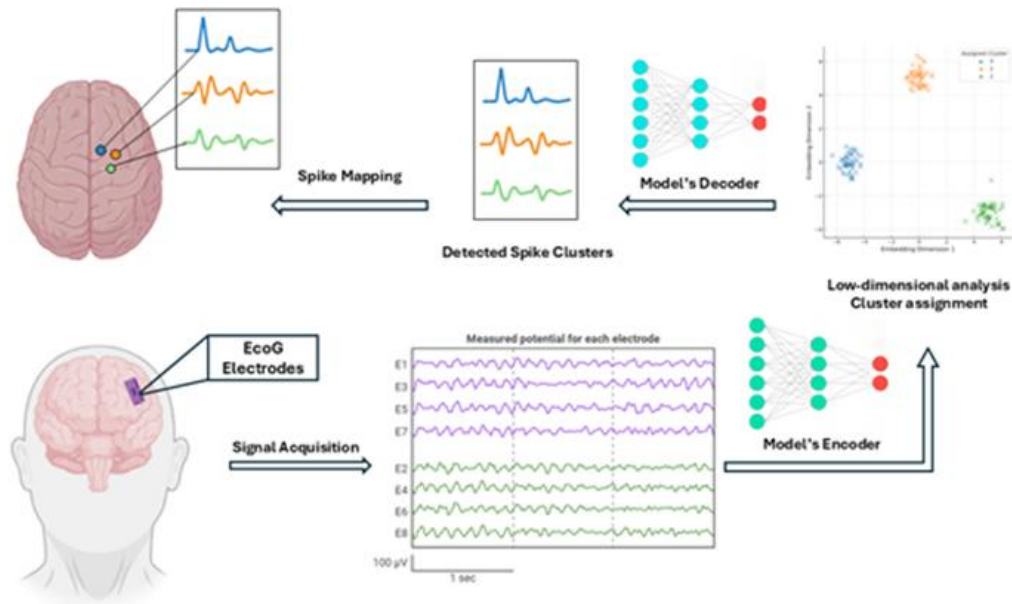


Figure 3. Deep learning-based pipeline for spike sorting from surface-level brain recordings. Surface neural signals are acquired from ECoG electrodes and preprocessed. The trained model's encoder transforms the signals into a low-dimensional embedding space, where spikes are clustered by similarity. The decoder assigns spikes to putative neuronal identities and reconstructs corresponding waveforms. Finally, spike clusters are mapped back to their estimated cortical locations.

References

1. Buzsáki, G., Anastassiou, C. A. & Koch, C. The origin of extracellular fields and currents — EEG, ECoG, LFP and spikes. *Nat. Rev. Neurosci.* **13**, 407–420 (2012).
2. Konerding, W. S. *et al.* New thin-film surface electrode array enables brain mapping with high spatial acuity in rodents. *Sci. Rep.* **8**, 3825 (2018).
3. Bockhorst, T. *et al.* Synchrony surfacing: epicortical recording of correlated action potentials. *Eur. J. Neurosci.* **48**, 3583–3596 (2018).
4. Einevoll, G. T., Franke, F., Hagen, E., Pouzat, C. & Harris, K. D. Towards reliable spike-train recordings from thousands of neurons with multielectrodes. *Curr. Opin. Neurobiol.* **22**, 11–17 (2012).
5. Rey, H. G., Pedreira, C. & Quiroga, R. Q. Past, present and future of spike sorting techniques. *Brain Res. Bull.* **119**, 106–117 (2015).
6. Yger, P. *et al.* A spike sorting toolbox for up to thousands of electrodes validated with ground truth simulation and in vitro experiments. *eLife* **7**, e34518 (2018).
7. Ray, S., Maunsell, J. H. R. & Treue, S. Neural correlates of high-gamma oscillations (60–200 Hz) in macaque local field potentials and their potential implications in electrocorticography. *J. Neurosci.* **28**, 11526–11536 (2008).
8. Manning, J. R. *et al.* Broadband shifts in local field potential power spectra are correlated with single-neuron spiking in humans. *J. Neurosci.* **29**, 13613–13620 (2009).

9. Ahmadi, N., Constandinou, T. G. & Bouganis, C. S. Inferring entire spiking activity from local field potentials. *Sci. Rep.* **11**, 19045 (2021).
10. Hong, N. *et al.* Machine learning-based high-frequency neuronal spike reconstruction from low-frequency and low-sampling-rate recordings. *Nat. Commun.* **15**, 635 (2024).
11. Swindale, N. V. & Spacek, M. A. Spike sorting for polytrodes: a divide and conquer approach. *Front. Syst. Neurosci.* **8**, 6 (2014).
12. Li, Z., Wang, Y., Zhang, N. & Li, X. An accurate and robust method for spike sorting based on convolutional neural networks. *Brain Sci.* **10**, 835 (2020).
13. Brockhoff, M. *et al.* PseudoSorter: a self-supervised spike sorting approach applied to reveal Tau-induced reductions in neuronal activity. *Sci. Adv.* **11**, eadr4155 (2025).
14. Lebedev, M. A. & Nicolelis, M. A. L. Brain–machine interfaces: from basic science to neuroprostheses and neurorehabilitation. *Physiol. Rev.* **97**, 767–837 (2017).
15. Jun, J. J. *et al.* Fully integrated silicon probes for high-density recording of neural activity. *Nature* **551**, 232–236 (2017).
16. Steinmetz, N. A. *et al.* Neuropixels 2.0: a miniaturized high-density probe for stable, long-term brain recordings. *Science* **372**, eabf4588 (2021).
17. Vansteensel, M. J. *et al.* Fully implanted brain–computer interface in a locked-in patient with ALS. *N. Engl. J. Med.* **375**, 2060–2066 (2016).
18. Dubey, A. & Ray, S. Cortical electrocorticogram (ECoG) is a local signal. *J. Neurosci.* **39**, 4299–4311 (2019).
19. Sampaio-Baptista, C. *et al.* fMRI neurofeedback in the motor system elicits bidirectional changes in white matter structure in the adult human brain. *Cell Rep.* **37**, 109890 (2021).
20. Logothetis, N. K., Kayser, C. & Oeltermann, A. In vivo measurement of cortical impedance spectrum in monkeys: implications for signal propagation. *Neuron* **55**, 809–823 (2007).
21. Schilling, A. *et al.* Deep learning-based decoding of single local field potential events. *NeuroImage* **297**, 120696 (2024).
22. Ardelean, E.-R. *et al.* A study of autoencoders as a feature extraction technique for spike sorting. *PLoS ONE* **18**, e0282810 (2023).
23. Livezey, J. A. & Glaser, J. I. Deep learning approaches for neural decoding across architectures and recording modalities. *Brief. Bioinform.* **22**, 1577–1591 (2021).
24. Bod, R. B. *et al.* From end to end: gaining, sorting, and employing high-density neural single unit recordings. *Front. Neuroinform.* **16**, 851024 (2022).
25. Abiri, R. *et al.* A comprehensive review of EEG-based brain–computer interface paradigms. *J. Neural Eng.* **16**, 011001 (2019).

PI's Approval



Azadeh Yazdan <azadehy@uw.edu>

To: Ali Alattar



Mon 6/9/2025 8:41 PM

Hi Ali,
Thank you for sending the proposal.
I approve it.
Best,
Azadeh

Azadeh Yazdan, PhD ([hear my name](#))
Associate Professor
Weill Neurohub Investigator
Department of Bioengineering
Department of Electrical and Computer Engineering
Washington National Primate Research Center
University of Washington
Phone: 206-543-6127

...